

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. The domain of the function $\sin^{-1}\left(\log_2\left(\frac{x}{3}\right)\right)$ is -
 (A) $[1, 3]$ (B) $\left[\frac{1}{2}, 3\right]$ (C) $\left[\frac{3}{2}, 6\right]$ (D) $\left[\frac{1}{2}, 2\right]$ MIT001
2. Domain of the function $f(x) = \log_e \cos^{-1}\{\sqrt{x}\}$ is, where $\{.\}$ represents fractional part function -
 (A) $x \in \mathbb{R}$ (B) $x \in [0, \infty)$ (C) $x \in (0, \infty)$ (D) $x \in \mathbb{R} - \{x \mid x \in \mathbb{I}\}$ MIT002
3. The range of the function $f(x) = \sin^{-1}(\log_2(-x^2 + 2x + 3))$ is -
 (A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{2}, 0\right]$ (C) $\left[0, \frac{\pi}{2}\right]$ (D) $[-1, 1]$ MIT003
4. Range of $f(x) = \cot^{-1}(\log_e(1 - x^2))$ is -
 (A) $(0, \pi)$ (B) $\left(0, \frac{\pi}{2}\right)$ (C) $\left[\frac{\pi}{2}, \pi\right)$ (D) $\left(0, \frac{\pi}{2}\right]$ MIT004
5. The function $f(x) = \cot^{-1} \sqrt{(x+3)x} + \cos^{-1} \sqrt{x^2 + 3x + 1}$ is defined on the set S , where S is equal to:
 (A) $\{0, 3\}$ (B) $(0, 3)$ (C) $\{0, -3\}$ (D) $[-3, 0]$ MIT005
6. Range of $f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$ is
 (A) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (B) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ (C) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$ (D) $\left(-\frac{\pi}{4}, \frac{3\pi}{4}\right)$ MIT006
7. $\operatorname{cosec}^{-1}(\cos x)$ exists if
 (A) $x \in [-1, 1]$ (B) $x \in \mathbb{R}$
 (C) x is an odd multiple of $\frac{\pi}{2}$ (D) x is a multiple of π MIT007
8. The value of $\cot^{-1} \left\{ \frac{\sqrt{1 - \sin x} + \sqrt{1 + \sin x}}{\sqrt{1 - \sin x} - \sqrt{1 + \sin x}} \right\}$, where $\frac{\pi}{2} < x < \pi$, is:
 (A) $\pi - \frac{x}{2}$ (B) $\frac{\pi}{2} + \frac{x}{2}$ (C) $\frac{x}{2}$ (D) $2\pi - \frac{x}{2}$ MIT008

9. The inequality $\sin^{-1}(\sin 5) > x^2 - 4x$ holds for
 (A) $x \in (2 - \sqrt{9 - 2\pi}, 2 + \sqrt{9 - 2\pi})$ (B) $x > 2 + \sqrt{9 - 2\pi}$
 (C) $x < 2 - \sqrt{9 - 2\pi}$ (D) $x \in \phi$
MIT009
10. Show that the roots r, s and t of the cubic $x(x - 2)(3x - 7) = 2$, are real and positive. Also compute the value of $\tan^{-1}(r) + \tan^{-1}(s) + \tan^{-1}(t)$.
 (A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) 2π
MIT010
11. The complete solution set of the inequality $[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \leq 0$, where $[.]$ denotes greatest integer function, is
 (A) $[-\infty, \cot 3]$ (B) $[\cot 3, \cot 2]$ (C) $[\cot 3, \infty)$ (D) $[-\infty, \cot 2]$
MIT011
12. If $2 \leq a < 3$, then the value of $\cos^{-1} \cos[a] + \operatorname{cosec}^{-1} \operatorname{cosec}[a] + \cot^{-1} \cot[a]$, (where $[.]$ denotes greatest integer less than equal to x) is equal to
 (A) $2 - \pi$ (B) $2 + \pi$ (C) π (D) 6
MIT012
13. $\tan^{-1}\left(1 - x^2 - \frac{1}{x^2}\right) + \sin^{-1}\left(x^2 + \frac{1}{x^2} - 1\right)$ (where $x \neq 0$) is equal to
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) π
MIT013
14. If $\cos^{-1}(2x^2 - 1) = 2\pi - 2\cos^{-1}x$, then -
 (A) $x \in [-1, 0]$ (B) $x \in [0, 1]$ (C) $x \in \left[0, \frac{1}{\sqrt{2}}\right]$ (D) $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$
MIT014
15. If $\cos^{-1}x - \cos^{-1}\frac{y}{2} = \alpha$, then $4x^2 - 4xy \cos \alpha + y^2$ is equal to-
 (A) $2 \sin 2\alpha$ (B) 4 (C) $4 \sin^2 \alpha$ (D) $-4 \sin^2 \alpha$
MIT015
16. $\tan\left(\frac{\pi}{4} + \frac{1}{2} \cos^{-1}x\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2} \cos^{-1}x\right)$, $x \neq 0$ is equal to
 (A) x (B) $2x$ (C) $\frac{2}{x}$ (D) $\frac{x}{2}$
MIT016
17. If $\alpha = 2 \arctan\left(\frac{1+x}{1-x}\right)$ & $\beta = \arcsin\left(\frac{1-x^2}{1+x^2}\right)$, what is the value of $\alpha + \beta$ if $x > 1$.
 (A) $-\pi$ (B) π (C) $\frac{\pi}{2}$ (D) 2π
MIT017

Inverse Trigonometric Functions

18. The numerical value of $\tan\left(2\tan^{-1}\frac{1}{5}-\frac{\pi}{4}\right)$ is

- (A) $\frac{-7}{17}$ (B) $\frac{7}{17}$ (C) $\frac{17}{7}$ (D) $-\frac{2}{3}$

MIT018

19. $\sum_{r=0}^{\infty} \tan^{-1}\left(\frac{r((r+1)!)}{(r+1)+((r+1)!)^2}\right)$ is equal to -

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\cot^{-1}3$ (D) $\tan^{-1}2$

MIT019

20. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \frac{2r+1}{r^4+2r^3+r^2+1}$ is equal to -

- (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $-\frac{\pi}{8}$

MIT020

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

- The values of 'x' satisfying the equation $\sin^{-1}(x^2 - 5x + 7) = 2\tan^{-1}1$ is (are)
 (A) 2 (B) 3 (C) 4 (D) 5
MIT047
- Consider the function $f(x) = e^x$ and $g(x) = \sin^{-1}x$, then which of the following is/are necessarily true.
 (A) Domain of $g \circ f =$ Domain of f (B) Range of $g \circ f \subset$ Range of g
 (C) Domain of $g \circ f$ is $(-\infty, 0]$ (D) Range of $g \circ f$ is $\left[-\frac{\pi}{2}, 0\right)$
MIT048
- Let $f(x) = \sin^{-1}(\tan x) + \cos^{-1}(\cot x)$ then
 (A) $f(x) = \frac{\pi}{2}$ wherever defined (B) domain of $f(x)$ is $x = n\pi \pm \frac{\pi}{4}, n \in I$
 (C) period of $f(x)$ is $\frac{\pi}{2}$ (D) $f(x)$ is many one function
MIT049
- Which of the following is/are correct ?
 (A) $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right) \forall x \in \mathbb{R} - \{0\}$
 (B) If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \operatorname{sgn}(e^x)$ then $f(x)$ is an into function.
 (C) If $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $f(x) = \sin x + x$ then $f(x)$ is an odd function.
 (D) If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \frac{e^x}{e^{[x]}}$ then $f(x)$ is a periodic function.
 (where $[.]$ represents greatest integer function)
MIT050
- If $\sin\left(2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right) + \cos\left(2\tan^{-1}\left(\frac{1}{3}\right)\right) = \frac{p}{q}$, where p & q are relatively prime then digit at units place of $(p - q)^{2k+1}, k \in \mathbb{N}$, can be -
 (A) 1 (B) 3 (C) 7 (D) 9
MIT051
- Value of $3\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ is greater than
 (A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) $\frac{5\pi}{6}$
MIT052
- If $\alpha = 2\tan^{-1}(\sqrt{3-2\sqrt{2}}) + \sin^{-1}\left(\frac{1}{\sqrt{6}-\sqrt{2}}\right)$, $\beta = \cot^{-1}(\sqrt{3}-2) + \frac{1}{8}\sec^{-1}(-2)$ & $\gamma = \tan^{-1}\frac{1}{\sqrt{2}} + \cos^{-1}\frac{1}{\sqrt{3}}$, then
 (A) $\alpha = \beta$ (B) $\alpha + \beta = 3\gamma$ (C) $4(\beta - \gamma) = \alpha$ (D) $\beta = \gamma$
MIT053

8. If $\sin^{-1}\left(\frac{\sqrt{x}}{2}\right) + \sin^{-1}\left(\sqrt{1-\frac{x}{4}}\right) + \tan^{-1}y = \frac{2\pi}{3}$, then:

(A) maximum value of $x^2 + y^2$ is $\frac{49}{3}$

(B) maximum value of $x^2 + y^2$ is 4

(C) minimum value of $x^2 + y^2$ is $\frac{1}{3}$

(D) minimum value of $x^2 + y^2$ is 3

MIT054

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 9 to 11

Consider a continuous function such that each image has atmost three preimage & atleast one image has exactly three preimages. This type of function is to be called as three-one function.

On the basis of above information, answer the following questions :

9. Which of the following function is a three-one function?

(A) $|\ln|x||$

(B) $e^{|x|}$

(C) $x^3 + 3x^2 - 7x + 6$

(D) $\cos(\cos^{-1}x)$

MIT055

10. If $f(x) = \sin^{-1}(\sin x)$ is a three – one function, then possible interval of x is -

(A) $[-\pi, \pi]$

(B) $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$

(C) $[-2\pi, 0]$

(D) $\left[\frac{-3\pi}{2}, \frac{\pi}{2}\right]$

MIT056

11. If $f(x)$ is a three-one function such that $f(a) = f(b)$ (where $a \neq b$), then number of maximum possible values of b is -

(A) 1

(B) 2

(C) 3

(D) 4

MIT057

Paragraph for Question 12 to 13

If $\alpha = 2\tan^{-1}\frac{1}{2} + \sin^{-1}\frac{3}{5}$, $\beta = \sin^{-1}\frac{12}{13} + \cos^{-1}\frac{4}{5} + \cot^{-1}\frac{16}{63}$.

On the basis of above information, answer the following questions :

12. If $2\sin\alpha, \cos\beta$ are roots of $x^2 - ax + b = 0$, then -

(A) $a = 1$

(B) $a = -1$

(C) $b = -2$

(D) $\frac{b}{a} = -2$

MIT058

13. The range of $\cot^{-1}(x^2 - 2x)$ is $(c, d]$, then

(A) $c = 0$

(B) $d = \frac{3\pi}{4}$

(C) $c = \frac{\pi}{4}$

(D) $c + d = \frac{3\pi}{4}$

MIT059

Paragraph for Question 14 and 15

Let $\alpha = \sin^{-1}x - \cos^{-1}y$ & $\frac{\pi}{2} - \alpha = \sin^{-1}y + \cos^{-1}x$.

14. $\cos^{-1}(-y)$ is equal to -

(A) $\frac{3\pi}{4}$

(B) $-\frac{\pi}{4}$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{2}$

MIT060

15. $[\alpha]$ can not be equal to (where $[.]$ denotes greatest integer function)

(A) -3

(B) -1

(C) 0

(D) 1

MIT061

MATCHING LIST TYPE QUESTION

Match List-I with List-II and select the correct answer.

List-I		List-II
(P) The number of integral values of k for which the equation $\sin^{-1} x^2 + \tan^{-1} x^2 = 2k + 1$ has a solution	(1)	1
(Q) Let $\sum_{k=1}^{\infty} \cot^{-1} \left(\frac{k^2}{8} \right) = \frac{p}{q} \pi$ where $\frac{p}{q}$ is rational in its lowest form, then find $ q - p $	(2)	2
(R) If $\tan^{-1} (\sin^2 \theta - 2 \sin \theta + 3) + \cot^{-1} (5 \sec^2 y + 1) = \frac{\pi}{2}$, then the value of $\cos^2 y - \sin \theta$	(3)	3
(S) Minimum positive integral value of x such that $f(x)$ is defined, $f(x) = \sqrt{\sec^{-1} \left(\frac{1- x }{2} \right)}$	(4)	0

16. Which of the following is the only **CORRECT** combination?

- (A) (P), (3) (B) (S), (2) (C) (R), (2) (D) (Q), (1)

MIT062

17. Which of the following is the only **CORRECT** combination?

- (A) (P), (1) (B) (Q), (1) (C) (R), (4) (D) (S), (2)

MIT063

MATRIX MATCH TYPE QUESTION

18. Match the column

Column-I		Column-II
(A) Let a, b, c be three positive real numbers $\theta = \tan^{-1} \sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1} \sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1} \sqrt{\frac{c(a+b+c)}{ab}}$, then θ is equal to	(p)	π
(B) The value of the expression $\tan^{-1} \left(\frac{1}{2} \tan 2A \right) + \tan^{-1} (\cot A) + \tan^{-1} (\cot^3 A)$ for $0 < A < (\pi/4)$ is equal to	(q)	$-\frac{\pi}{2}$
(C) If $x < 0$, then $\frac{1}{2} \{ \cos^{-1}(2x^2 - 1) + 2 \cos^{-1} x \}$ is equal to	(r)	$-\pi$
(D) The value of $\sin^{-1} \left(\frac{3}{5} \right) - \cos^{-1} \left(\frac{12}{13} \right) + \cos^{-1} \left(\frac{16}{65} \right)$ is equal to	(s)	$\frac{\pi}{2}$

MIT064

MIT065

MIT066

MIT067

Inverse Trigonometric Functions

19. Match the column :

Column-I	Column-II
(P) $\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{8}{19}\right)$	(1) $\tan^{-1}\left(\frac{1}{2}\right)$
(Q) $\tan^{-1}\left(\frac{2}{11}\right) + \tan^{-1}\left(\frac{7}{24}\right)$	(2) $\frac{\pi}{4}$
(R) $2\cos^{-1}\frac{3}{\sqrt{13}} + \cot^{-1}\frac{16}{63} + \frac{1}{2}\cos^{-1}\frac{7}{25}$	(3) $\frac{\pi}{6}$
(S) $\arccos\sqrt{\frac{2}{3}} - \arccos\frac{\sqrt{6}+1}{2\sqrt{3}}$	(4) π

MIT087

20. Solve for x :

Column-I	Column-II
(P) $\sin^{-1}x + \sin^{-1}2x = \frac{\pi}{3}$	(1) $x = 3$
(Q) $\tan^{-1}\frac{1}{1+2x} + \tan^{-1}\frac{1}{1+4x} = \tan^{-1}\frac{2}{x^2}$	(2) $x = \frac{1}{2}\sqrt{\frac{3}{7}}$
(R) $\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$	(3) $\left[\frac{\sqrt{3}}{2}, 1\right]$
(S) $3\cos^{-1}x = \sin^{-1}\left(\sqrt{1-x^2}(4x^2-1)\right)$	(4) $x = 0, \frac{1}{2}, -\frac{1}{2}$

MIT088

EXERCISE (S)

1. The value of $\tan^2(\sec^{-1} 3) + \cot^2(\operatorname{cosec}^{-1} 4)$ is - MIT022
2. If $\sin^{-1}\sin\left(\frac{10\pi}{3}\right) + \cos^{-1}\cos\frac{22\pi}{3} + \tan^{-1}\tan 10 = a\pi + b$, then $\left|\frac{3ab}{80}\right|$ is equal to MIT030
3. The value of $\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right\}$ is- MIT025
4. If the value of $\cot\left(\operatorname{cosec}^{-1}\frac{5}{3} + \tan^{-1}\frac{2}{3}\right)$ is equal to $\frac{a}{b}$, where a and b are coprime, then the value of $a + b$ is MIT091
5. Number of solution(s) of the equation $\cos^{-1}\sqrt{x} - \sin^{-1}\sqrt{x-1} + \cos^{-1}\sqrt{1-x} - \sin^{-1}\frac{1}{\sqrt{x}} = \frac{\pi}{2}$ is - MIT026
6. The number of solution(s) of the equation, $\sin^{-1}x + \cos^{-1}(1-x) = \sin^{-1}(-x)$, is MIT027
7. Let $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, then the value of $f(1) + f(2)$ is $k\pi$ where k is equal to MIT021
8. If $x > 0$ $\cos^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2} - \cos^{-1}\left(\frac{16}{x}\right)$ then x is - MIT023
9. Number of integral ordered pairs (a, b) for which $\sin^{-1}(1+b+b^2+\dots\infty) + \cos^{-1}\left(a - \frac{a^2}{3} + \frac{a^3}{9} - \dots\infty\right) = \frac{\pi}{2}$ is - MIT024
10. If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots\right) = \frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$, then x equals MIT028
11. Let $y = \sin^{-1}(\sin 8) - \tan^{-1}(\tan 10) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 7)$. If y simplifies to $a\pi + b$, then find $(a-b)$. MIT029
12. If the domain of definition of the function $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ for real valued ' x ' is $[\alpha, \beta]$, then the value of $4(\alpha + \beta)$ is equal to: MIT068
13. Number of solutions of equation $\tan^{-1}(e^{-x}) + \cot^{-1}(|\ell nx|) = \pi/2$ is : MIT069
14. If $\tan^{-1}x + \cot^{-1}\frac{1}{y} + 2\tan^{-1}z = \pi$, then $x + y + \mu z = xz^2 + yz^2 + \lambda xyz$. Find $\lambda + \mu$ MIT089
15. If $a > 0, b > 0$, $\tan^{-1}\left(\frac{a}{x}\right) + \tan^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$ then $x = (ab)^{\frac{1}{n}}$ find $n(n \in \mathbb{N})$ MIT090

EXERCISE - JEE (Main) PYQ

1. If $\cos^{-1}\left(\frac{2}{3x}\right) + \cos^{-1}\left(\frac{3}{4x}\right) = \frac{\pi}{2}$, $\left(x > \frac{3}{4}\right)$, then x is equal to : [JEE (Main) 2019]

(1) $\frac{\sqrt{145}}{12}$ (2) $\frac{\sqrt{145}}{10}$ (3) $\frac{\sqrt{146}}{12}$ (4) $\frac{\sqrt{145}}{11}$

MIT031

2. If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to: [JEE (Main) 2019]

(1) π (2) 7π (3) 0 (4) 10

MIT032

3. The value of $\cot\left(\sum_{n=1}^{19} \cot^{-1}\left(1 + \sum_{p=1}^n 2p\right)\right)$ is : [JEE (Main) 2019]

(1) $\frac{22}{23}$ (2) $\frac{23}{22}$ (3) $\frac{21}{19}$ (4) $\frac{19}{21}$

MIT033

4. Considering only the principal values of inverse functions, the set

$$A = \left\{x \geq 0 : \tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}\right\}$$

[JEE (Main) 2019]

- (1) is an empty set (2) Contains more than two elements
(3) Contains two elements (4) is a singleton

MIT034

5. The value of $\sin^{-1}\left(\frac{12}{13}\right) - \sin^{-1}\left(\frac{3}{5}\right)$ is equal to : [JEE (Main) 2019]

(1) $\pi - \sin^{-1}\left(\frac{63}{65}\right)$ (2) $\pi - \cos^{-1}\left(\frac{33}{65}\right)$ (3) $\frac{\pi}{2} - \sin^{-1}\left(\frac{56}{65}\right)$ (4) $\frac{\pi}{2} - \cos^{-1}\left(\frac{9}{65}\right)$

MIT037

6. If S is the sum of the first 10 terms of the series $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + \tan^{-1}\left(\frac{1}{21}\right) + \dots$, then $\tan(S)$ is equal to : [JEE (Main) 2020]

(1) $\frac{5}{11}$ (2) $-\frac{6}{5}$ (3) $\frac{10}{11}$ (4) $\frac{5}{6}$

MIT038

7. The domain of the function $f(x) = \sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$ is $(-\infty, -a] \cup [a, \infty)$. Then a is equal to : [JEE (Main) 2020]

(1) $\frac{1+\sqrt{17}}{2}$ (2) $\frac{\sqrt{17}-1}{2}$ (3) $\frac{\sqrt{17}}{2}+1$ (4) $\frac{\sqrt{17}}{2}$

MIT092

8. $2\pi - \left(\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{16}{65}\right)$ is equal to: [JEE (Main) 2020]

(1) $\frac{7\pi}{4}$ (2) $\frac{5\pi}{4}$ (3) $\frac{3\pi}{2}$ (4) $\frac{\pi}{2}$

MIT093

9. Let M and m respectively be the maximum and minimum values of the function $f(x) = \tan^{-1}(\sin x + \cos x)$ in $\left[0, \frac{\pi}{2}\right]$. Then the value of $\tan(M-m)$ is equal to: [JEE (Main) 2021]
 (1) $2 + \sqrt{3}$ (2) $2 - \sqrt{3}$ (3) $3 + 2\sqrt{2}$ (4) $3 - 2\sqrt{2}$ MIT039
10. If $\sum_{r=1}^{50} \tan^{-1} \frac{1}{2r^2} = p$, then the value of $\tan p$ is: [JEE (Main) 2021]
 (1) $\frac{101}{102}$ (2) $\frac{50}{51}$ (3) 100 (4) $\frac{51}{50}$ MIT040
11. If $0 < a, b < 1$, and $\tan^{-1}a + \tan^{-1}b = \frac{\pi}{4}$, then the value of $(a+b) - \left(\frac{a^2+b^2}{2}\right) + \left(\frac{a^3+b^3}{3}\right) - \left(\frac{a^4+b^4}{4}\right) + \dots$ is: [JEE (Main) 2021]
 (1) $\log_e 2$ (2) $e^2 - 1$ (3) e (4) $\log_e \left(\frac{e}{2}\right)$ MIT041
12. The domain of the function $f(x) = \sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right) + \cos^{-1}\left(\frac{x-1}{x+1}\right)$ is: [JEE (Main) 2021]
 (1) $\left[0, \frac{1}{4}\right]$ (2) $[-2, 0] \cup \left[\frac{1}{4}, \frac{1}{2}\right]$ (3) $\left[\frac{1}{4}, \frac{1}{2}\right] \cup \{0\}$ (4) $\left[0, \frac{1}{2}\right]$ MIT094
13. $\operatorname{cosec}\left[2\cot^{-1}(5) + \cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to: [JEE (Main) 2021]
 (1) $\frac{56}{33}$ (2) $\frac{65}{56}$ (3) $\frac{65}{33}$ (4) $\frac{75}{56}$ MIT095
14. The domain of the function $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ is: [JEE (Main) 2022]
 (1) $R - \left\{-\frac{1}{2}, \frac{1}{2}\right\}$ (2) $(-\infty, -1] \cup [1, \infty) \cup \{0\}$
 (3) $(-\infty, \frac{-1}{2}) \cup (\frac{1}{2}, \infty) \cup \{0\}$ (4) $(-\infty, \frac{-1}{\sqrt{2}}] \cup [\frac{1}{\sqrt{2}}, \infty) \cup \{0\}$ MIT042
15. The set of all values of k for which $(\tan^{-1}x)^3 + (\cot^{-1}x)^3 = k\pi^3, x \in R$, is the interval: [JEE (Main) 2022]
 (1) $\left[\frac{1}{32}, \frac{7}{8}\right]$ (2) $\left(\frac{1}{24}, \frac{13}{16}\right)$ (3) $\left[\frac{1}{48}, \frac{13}{16}\right]$ (4) $\left[\frac{1}{32}, \frac{9}{8}\right]$ MIT043
16. If the inverse trigonometric functions take principal values, then $\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right) + \frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal to: [JEE (Main) 2022]
 (1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{6}$ MIT045

17. The domain of the function $f(x) = \frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e(x^2-3x+2)}$ is [JEE (Main) 2022]
 (1) $(-\infty, 1) \cup (2, \infty)$ (2) $(2, \infty)$
 (3) $\left[-\frac{1}{2}, 1\right) \cup (2, \infty)$ (4) $\left[-\frac{1}{2}, 1\right) \cup (2, \infty) - \left\{\frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}\right\}$
 MIT046
18. Let $f(x) = 2\cos^{-1}x + 4\cot^{-1}x - 3x^2 - 2x + 10$, $x \in [-1, 1]$. If $[a, b]$ is the range of the function then $4a - b$ is equal to: [JEE (Main) 2022]
 (1) 11 (2) $11 - \pi$ (3) $11 + \pi$ (4) $15 - \pi$
 MIT096
19. $50 \tan\left(3\tan^{-1}\left(\frac{1}{2}\right) + 2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right) + 4\sqrt{2} \tan\left(\frac{1}{2}\tan^{-1}(2\sqrt{2})\right)$ is equal to. [JEE (Main) 2022]
 MIT097
20. The range of $f(x) = 4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is [JEE (Main) 2023]
 (1) $[0, \pi]$ (2) $[0, 2\pi]$ (3) $[0, \pi)$ (4) $[0, 2\pi)$
 MIT081
21. $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to [JEE (Main) 2023]
 (1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{6}$
 MIT082
22. If the sum of all the solutions of $\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \cot^{-1}\left(\frac{1-x^2}{2x}\right) = \frac{\pi}{3}$, $-1 < x < 1, x \neq 0$, is $\alpha - \frac{4}{\sqrt{3}}$, then α is equal to. [JEE (Main) 2023]
 MIT083
23. Let $a_1 = 1, a_2, a_3, a_4, \dots$ be consecutive natural numbers. Then $\tan^{-1}\left(\frac{1}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{1}{1+a_2a_3}\right) + \dots + \tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ is equal to [JEE (Main) 2023]
 (1) $\frac{\pi}{4} - \cot^{-1}(2022)$ (2) $\cot^{-1}(2022) - \frac{\pi}{4}$ (3) $\tan^{-1}(2022) - \frac{\pi}{4}$ (4) $\frac{\pi}{4} - \tan^{-1}(2022)$
 MIT084
24. Let $(a, b) \subset (0, 2\pi)$ be the largest interval for which $\sin^{-1}(\sin\theta) - \cos^{-1}(\sin\theta) > 0$, $\theta \in (0, 2\pi)$, holds. If $\alpha x^2 + \beta x + \sin^{-1}(x^2 - 6x + 10) + \cos^{-1}(x^2 - 6x + 10) = 0$ and $\alpha - \beta = b - a$, then α is equal to: [JEE (Main) 2023]
 (1) $\frac{\pi}{48}$ (2) $\frac{\pi}{16}$ (3) $\frac{\pi}{8}$ (4) $\frac{\pi}{12}$
 MIT085

25. If $\sin^{-1} \frac{\alpha}{17} + \cos^{-1} \frac{4}{5} - \tan^{-1} \frac{77}{36} = 0$, $0 < \alpha < 13$, then $\sin^{-1}(\sin \alpha) + \cos^{-1}(\cos \alpha)$ is equal to
 (1) π (2) 16 (3) 0 (4) $16 - 5\pi$ [JEE (Main) 2023] MIT086
26. Given the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in $[-1, 1]$ such that $\cos^{-1} x - \sin^{-1} y = \alpha$, $-\frac{\pi}{2} \leq \alpha \leq \pi$. Then, the minimum value of $x^2 + y^2 + 2xy \sin \alpha$ is
 (1) -1 (2) 0 (3) $-\frac{1}{2}$ (4) $\frac{1}{2}$ [JEE (Main) 2024] MIT148
27. For $n \in N$, if $\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^{-1} n = \frac{\pi}{4}$, then n is equal to _____. [JEE (Main) 2024] MIT149
28. If the domain of the function $f(x) = \sin^{-1} \left(\frac{x-1}{2x+3} \right)$ is $R - (\alpha, \beta)$ then $12\alpha\beta$ is equal to:
 (1) 36 (2) 24 (3) 40 (4) 32 [JEE (Main) 2024] MIT150
29. Let $x = \frac{m}{n}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos(2\sin^{-1} x) = \frac{1}{9}$ and let α, β ($\alpha > \beta$) be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point (α, β) lies on the line
 (1) $3x + 2y = 2$ (2) $5x - 8y = -9$ (3) $3x - 2y = -2$ (4) $5x + 8y = 9$ [JEE (Main) 2024] MIT151
30. If $a = \sin^{-1}(\sin(5))$ and $b = \cos^{-1}(\cos(5))$, then $a^2 + b^2$ is equal to [JEE (Main) 2024] MIT152
 (1) $4\pi^2 + 25$ (2) $8\pi^2 - 40\pi + 50$ (3) $4\pi^2 - 20\pi + 50$ (4) 25

EXERCISE - JEE (Advanced) PYQ

1. The value of $\cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \sum_{k=1}^n 2k\right)\right)$ is [JEE (Advanced) 2013]

(A) $\frac{23}{25}$ (B) $\frac{25}{23}$ (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

MIT070

2. Match List-I with List-II and select the correct answer using the code given below the lists.

List-I

List-II

P. $\left(\frac{1}{y^2} \left(\frac{\cos(\tan^{-1} y) + y \sin(\tan^{-1} y)}{\cot(\sin^{-1} y) + \tan(\sin^{-1} y)}\right)^2 + y^4\right)^{1/2}$ takes value 1. $\frac{1}{2}\sqrt{5}$

Q. If $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$,
then possible value of $\cos \frac{x-y}{2}$ is 2. $\sqrt{2}$

R. If $\cos\left(\frac{\pi}{4} - x\right) \cos 2x + \sin x \sin 2x \sec x$ 3. $\frac{1}{2}$
 $= \cos x \sin 2x \sec x + \cos\left(\frac{\pi}{4} + x\right) \cos 2x$,

then possible value of $\sec x$ is

S. If $\cot(\sin^{-1} \sqrt{1-x^2}) = \sin(\tan^{-1}(x\sqrt{6}))$, $x \neq 0$, 4. 1

then possible value of x is

[JEE (Advanced) 2013]

Codes :

	P	Q	R	S
(A)	4	3	1	2
(B)	4	3	2	1
(C)	3	4	2	1
(D)	3	4	1	2

MIT071

3. Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is [JEE (Advanced) 2014]

MIT072

4. If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$ where the inverse trigonometric functions take only the principal values, then the correct option(s) is(are) [JEE (Advanced) 2015]

(A) $\cos \beta > 0$ (B) $\sin \beta < 0$ (C) $\cos(\alpha + \beta) > 0$ (D) $\cos \alpha < 0$

MIT073

5. The number of real solutions of the equation

$$\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty} \left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty} (-x)^i\right) \text{ lying in the interval } \left(-\frac{1}{2}, \frac{1}{2}\right) \text{ is } ______$$

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume value in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.)

[JEE (Advanced) 2018]

MIT074

6. Let $E_1 = \left\{x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0\right\}$ and $E_2 = \left\{x \in E_1 : \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right) \text{ is a real number}\right\}$.

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.)

Let $f : E_1 \rightarrow \mathbb{R}$ be the function defined by $f(x) = \log_e\left(\frac{x}{x-1}\right)$

and $g : E_2 \rightarrow \mathbb{R}$ be the function defined by $g(x) = \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$.

LIST-I

- P. The range of f is
- Q. The range of g contains
- R. The domain of f contains
- S. The domain of g is

LIST-II

1. $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
2. $(0, 1)$
3. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
4. $(-\infty, 0) \cup (0, \infty)$
5. $\left(-\infty, \frac{e}{e-1}\right]$
6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1}\right]$

The correct option is :

[JEE (Advanced) 2018]

(A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 1$

(B) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 6; S \rightarrow 5$

(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 6$

(D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 6; S \rightarrow 5$

MIT075

7. The value of $\sec^{-1}\left(\frac{1}{4} \sum_{k=0}^{10} \sec\left(\frac{7\pi}{12} + \frac{k\pi}{2}\right) \sec\left(\frac{7\pi}{12} + \frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ equals

[JEE (Advanced) 2019]

MIT076

8. For any positive integer n , let $S_n : (0, \infty) \rightarrow \mathbb{R}$ be defined by $S_n(x) = \sum_{k=1}^n \cot^{-1} \left(\frac{1+k(k+1)x^2}{x} \right)$, where

for any $x \in \mathbb{R}$, $\cot^{-1}x \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is

(are) **TRUE** ?

[JEE (Advanced) 2021]

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1} \left(\frac{1+11x^2}{10x} \right)$, for all $x > 0$

(B) $\lim_{n \rightarrow \infty} \cot(S_n(x)) = x$, for all $x > 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \leq \frac{1}{2}$, for all $n \geq 1$ and $x > 0$

MIT077

9. Considering only the principal values of the inverse trigonometric functions, the value of

$\frac{3}{2} \cos^{-1} \sqrt{\frac{2}{2+\pi^2}} + \frac{1}{4} \sin^{-1} \frac{2\sqrt{2}\pi}{2+\pi^2} + \tan^{-1} \frac{\sqrt{2}}{\pi}$ is _____.

[JEE (Advanced) 2022]

MIT078

10. Let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, for $x \in \mathbb{R}$. Then the number of real solutions of the equation

$\sqrt{1+\cos(2x)} = \sqrt{2} \tan^{-1}(\tan x)$ in the set $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ is equal to

[JEE (Advanced) 2023]

MIT079

11. For any $y \in \mathbb{R}$, let $\cot^{-1}(y) \in (0, \pi)$ and $\tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the sum of all the solutions of

the equation $\tan^{-1} \left(\frac{6y}{9-y^2} \right) + \cot^{-1} \left(\frac{9-y^2}{6y} \right) = \frac{2\pi}{3}$ for $0 < |y| < 3$, is equal to

[JEE (Advanced) 2023]

(A) $2\sqrt{3}-3$

(B) $3-2\sqrt{3}$

(C) $4\sqrt{3}-6$

(D) $6-4\sqrt{3}$

MIT080

12. Considering only the principal values of the inverse trigonometric functions, the value of

$\tan \left(\sin^{-1} \left(\frac{3}{5} \right) - 2 \cos^{-1} \left(\frac{2}{\sqrt{5}} \right) \right)$ is

[JEE (Advanced) 2024]

(A) $\frac{7}{24}$

(B) $\frac{-7}{24}$

(C) $\frac{-5}{24}$

(D) $\frac{5}{24}$

MIT153

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	A	C	C	C	D	B	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	B	A	C	C	A	A	B	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	B,C	A,B,C,D	B,D	B,C	A,B	A,C	A,C	C	A
Que.	11	12	13	14	15	16	17	18		
Ans.	B	A,C,D	A,B,D	A	D	C	A	$A \rightarrow p ; B \rightarrow p ; C \rightarrow p ; D \rightarrow s$		
Que.	19				20					
Ans.	$P \rightarrow 2 ; Q \rightarrow 1 ; R \rightarrow 4 ; S \rightarrow 3$				$P \rightarrow 2 ; Q \rightarrow 1 ; R \rightarrow 4 ; S \rightarrow 3$					

EXERCISE (S)

1.	23	2.	1	3.	0	4.	23	5.	0
6.	1	7.	2	8.	20	9.	0	10.	1
11.	53	12.	1	13.	2	14.	4	15.	2

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	3	4	3	4	1	3	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	2	4	1	3	4	2	29	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	1,3	4	1	2	47	4	4	2

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	3	B,C,D	2	A	0.00	A,B	2.35 or 2.36	3
Que.	11	12								
Ans.	C	B								